Module 13 Lecture - F Distribution and One-Way ANOVA

Introduction to Statistical Methods

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Interpret the F probability distribution as the number of groups and the sample size change.
- Discuss one use for the F distribution: one-way ANOVA
- Conduct and interpret one-way ANOVA.

1.2 Instructor Learning Objectives

- Appreciate the ANOVA as another test in your toolbox alongside the other tests for dealing with categorical and continuous outcome variables
- Understand the F-distribution as another practical distribution for deriving inferences and conclusions

1.3 Introduction

Discuss: When we needed to compare the averages of two groups on a continuous variable, what statistical test would we use?

- While we've already covered how to test ______ between two groups, now we need to consider the scenario with ______ or more groups on some outcome
  - Examples:
    - Comparing class outcomes for those in hybrid, in-person, or online sections
    - Comparing students on three different tracks in high school: regular, advanced, or remedial
- When we are comparing three ______ groups on some numeric, ______ outcome, we will employ the one-way ANOVA, standing for Analysis of Variance (ANOVA)
  - ANOVA is a ______ family of techniques with many applications and extensions

- We are looking at the one-way ANOVA as a ____________ use of this technique

> **Important**
>
> This is very similar to how we introduced simple linear regression - there are many more advancements on top of what you learn in this class, which you will encounter in more advanced statistics classes!

- Why not use a bunch of ____________ across the 3 groups instead of this new complicated method?
  - Problem: It raises our Type I error rate
  - This is something done as part of ANOVA in ____________ testing, but that won't be covered here

> **Discuss:** Describe what Type I errors are

- Our previous tests all made use of specific, practical ____________ for determining significance:
  - z-tests $\longrightarrow$ normal distribution
  - t-tests $\longrightarrow$ t-distribution
  - χ^2 tests $\longrightarrow$ χ^2 distribution
  - The ANOVA family will introduce a new distribution: the ____________
- We'll start by describing the F-distribution itself, then the F-ratio/statistic that we compute, and then finally the application within the One-way ANOVA

2 The F Distribution

2.1 Introduction

- The F-distribution was developed by Sir Ronald Fisher and is a unique distribution applied often for the ____________ family of statistical tests
  - In ____________ it is given as $F \sim F_{df(num),df(denom)}$ where:
  - $df(num) \rightarrow df_{between}$ and $df(denom) \rightarrow df_{within}$
  - Example: $F \sim F_{2,24}$ has:

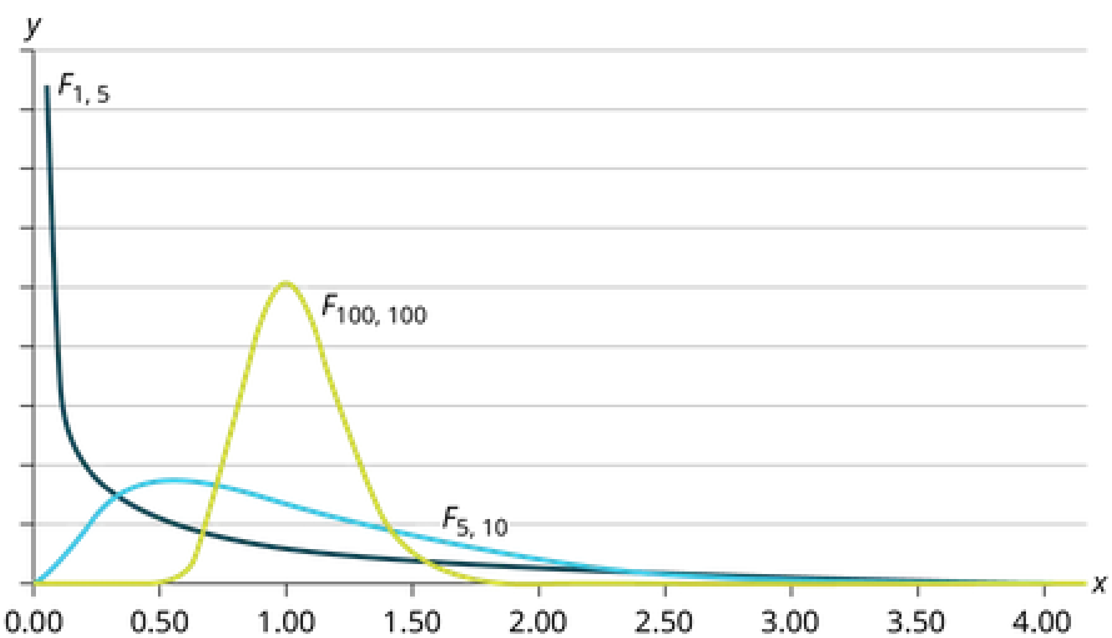
- $df_{between} = 2$
- $df_{within} = 24$

> **Important**
>
> Hold on just a second, *two* different degrees of freedom? Yes! We'll discuss more about why that is in our discussion on the F-ratio itself

- The F-distribution actually consists of values that are squares of the t-distribution, and is derived from the t-distribution
  - How exactly Fisher proved that is *way* beyond the scope of this class

> Discuss: Recall, what exactly is the distinction that makes the t-distribution different and useful over the normal distribution? The F-statistic benefits from this the same!

2.2 Additional Facts About the F Distribution



- The curve is not ________________ but skewed to the right.

  - Thus, it has a more similar look to the ______ distribution, compared to the t-distribution
- There is a different curve for each set of ______.
  - Once again, similar to chi-squared
- The F statistic is greater than or equal to zero.

> Discuss: After you work through the F-ratio section and calculation below, return to this section and try to explain, using the formula, why this F is always greater than or equal to zero

- As the degrees of freedom for the numerator and for the denominator get larger, the curve approximates the ______ distribution.
  - Remember, the $df_{between} \rightarrow$ numerator and the $df_{within} \rightarrow$ denominator

3 The F Ratio

3.1 Introduction

- The F-ratio is just another name for the F statistic that ______ from the test we run
  - The F-ratio/statistic we get is a ratio of:
    * The variances ______ samples
    * The variances ______ samples
    * It's formula is given as:

$$F = \frac{MS_{between}}{MS_{within}}$$

- Well describe what the MS means here in a second

3.2 Between & Within Variation

- For Variances between samples, we are considering how much __________ there is between the 3 or more groups we are comparing to one another
  - This may also be known as “variation due to __________” or “__________ variation”
  - This will be represented __________ as the $MS_{between}$ or the mean square
  - This is the __________ of the F-ratio formula
- In the case of Variances within samples, we example how much variation there is within __________ of the groups we are comparing to one another
  - This may also be known as “variation due to __________” or “__________ variation”
  - This will be represented as MS_{within} in the following formula
  - This is the __________ of the F-ratio formula

3.3 Calculation of the F-ratio/statistic

- We are *not* going to calculate F by hand, as it is very time-consuming, albeit plenty possible
  - However, you should still follow along and make sure you can see the flow of information as we substitute numbers in these calculations

3.3.1 Overall F Calculation

$$F = \frac{MS_{between}}{MS_{within}}$$

3.3.2 Between-group Calculations

$$MS_{between} = \frac{SS_{between}}{df_{between}}$$

$$df_{between} = k - 1$$

$$SS_{between} = \sum [\frac{(s_j)^2}{n_j}] - \frac{(\sum s_j)^2}{n}$$

$$SS_{total} = \sum x^2 \cdot \frac{(\sum x)^2}{n}$$

- Where
  - $SS_{between}$: Sum of squares between groups
  - SS_{total} : Sum of squares total
  - $df_{between}$: degrees of freedom between groups
  - k : the number of groups
  - n : total sample size
  - n_j : the size of j^{th} group
  - s_j : sum of values of j^{th} group

3.3.3 Within-group Calculations

$$MS_{within} = \frac{SS_{within}}{df_{within}}$$

$$df_{within} = n - k$$

$$SS_{within} = SS_{total} - SS_{between}$$

- Where
  - SS_{within} : Sum of squares between groups
  - $SS_{between}$: Sum of squares between groups
  - SS_{total} : Sum of squares total
  - df_{within} : degrees of freedom between groups
  - n : total sample size

3.4 Conclusions with the F-ratio/statistic

Discuss: Review: describe the general processes of hypothesis testing, starting with setting the null and alternative hypotheses and ending with a rejection or retaining of the null hypothesis. Try and use the words 'rare event' at some point in your explanation.

- The F statistic, which we derive from the formula for whatever test we are using, is actually best described as a __________ or fraction (we'll return to this ratio/statistic in a bit!)
  - Much like previous test statistics, we use this to determine if our results are __________ enough to describe our results as unlikely if the null hypothesis is __________
  - As usual, the F statistic has a corresponding __________ that tell us the probability of results due to chance under the null hypothesis

? What value do we compare p against to determine if we can reject the null hypothesis?

- (●) A) Alpha
- ( ) B) Omega
- ( ) C) Confidence Level
- ( ) D) Beta

Explanation:

- For __________ testing of the F-ratio, our one-way ANOVA test will always be b58fc729-690b-4000-b19f-365a4093b2ff-7B7B3C20626C616E6B207269676874203E7D7D–tailed due to the ratio nature of statistic, with larger numbers suggest greater variation between groups relative to the variation within groups!
  - Put another way, we are trying to see if it is __________ enough out on the right tail to say that it is significant!

Discuss: Which other distribution also always had a right-tailed application? Explain why that is using the relevant formula for that statistic.

4 One-way ANOVA

4.1 Introduction

Important

In the prior calculations, we covered the mathematical computation in a one-way ANOVA, but this next part will cover the more conceptual side of things.

- The goal of the one-way ANOVA is to determine if there are __________ differences between __________ group means
  - This is done by examining the __________ of the groups (as we previously discussed as part of The F Ratio)
- This is how we will practically use the F-distribution

4.2 Assumptions

- Much like the other tests, we need to be mindful of several __________ that underline this statistical test
- These assumptions are:
  - Each population from which a sample is taken is assumed to be __________.
  - All samples are __________ selected and independent.
  - The populations are assumed to have __________ standard deviations (or variances).
  - The factor is a __________ variable.

  - The response is a ________ variable.

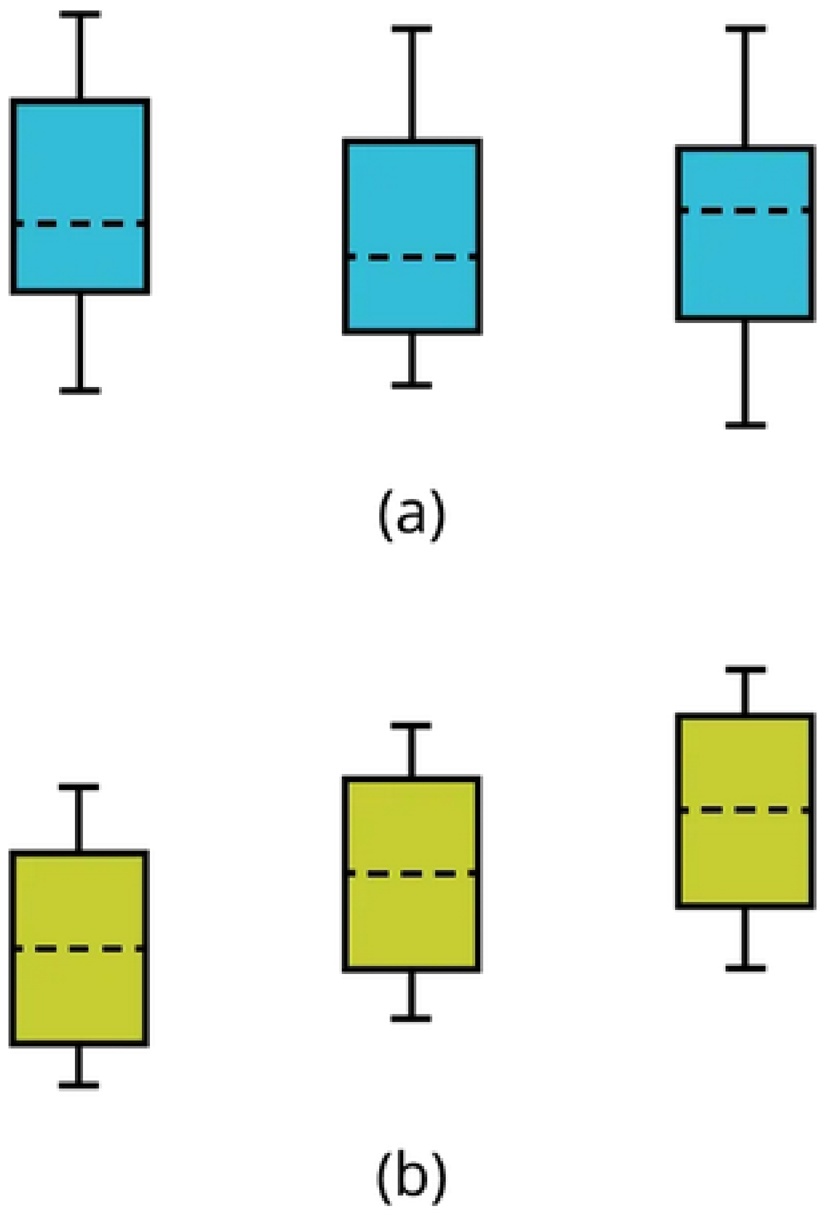
4.3 Null and Alternative Hypotheses

- In a one-way ANOVA, the ________ hypothesis is as follows:
  - $H_0 : \mu_1 = \mu_2 = \mu_3 ... = \mu_k$
  - H_A : At least two of the group means $\mu_1, \mu_2, \mu_3, ..., \mu_k$ are not equal

> **Important**
>
> Be *very* careful in how you phrase the alternative hypothesis for this test, it looks a little bit different than the other tests we've covered before!

- Effectively, a statistically significant F-statistic only tells us that a difference exists ________, but not where that difference lies
  - More explanation on this in the Extensions section!

> **Discuss:** What is the name of type of the following plots?



4.4 Extensions

- Of course, this is only the tip of the iceberg
- One question that the ANOVA alone __________ answer: which of the groups are different?
  - If we have a __________ one-way ANOVA test, then we can follow up with post-hoc testing, which can tell us which __________ exactly are different than one another

- There is also the two-way ANOVA, which allows us to compare two or more ________ variables and their combined and interacting impact on a single dependent variable in two-way Analysis of Variance.
  - While beyond the scope of this class, ________ are an incredibly important part of a lot of more complicated studies
- Finally, there is the repeated-measures ANOVA

> Discuss: Which of the previous tests we talked about dealt with comparing two sets of measurements, taken at two separate time points, on the same sample?

5 Conclusion

5.1 Recap

- We can think of a one-way ANOVA as being similar to an independent-samples t-test, but for more than 2 groups. Just note the differences in the null/alternative hypotheses setup
- There are some important extensions and other applications of the ANOVA that we did not comprehensively cover here. However, they do still employ the F-distribution and have the underlying focus on comparing variances to make conclusions about differences between groups
- The F-distribution adds to the other practical distributions employed as part of hypothesis testing, such as the t-distribution and chi-squared distribution. Just like those, it allows us to conclude whether a null hypothesis can be rejected or retained.

Key Terms

A

Analysis of Variance (ANOVA) A family of statistical tests used to compare variances as a way to detect differences between more than two groups 2

F

F-distribution A distribution developed by Ronald Fisher, expressed as a ratio, used commonly for the ANOVA family of tests 3
F-ratio Another name for the F-statistic derived from an ANOVA test, based upon the ratio of the variances between the samples to the variances within the samples 5

M

mean square The computational representation of variation between groups 6

P

post-hoc testing Statistical testing done after a significant ANOVA to tease apart where exactly differences lie between the groups 11

R

repeated-measures ANOVA A type of ANOVA used when measurements are taken on the same sample 3 or more times, to compare against each other at different time points 12

T

two-way ANOVA A type of ANOVA that allows for testing two independent (grouping) variables impact on a single dependent (outcome) variable 12

V

Variances between samples How much differences there are between the 3 or more groups that we are comparing 6
Variances within samples How much variances occur within each of the groups we are comparing against one another 6

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms